

7. Acquisition C: disk and cutout area series

Each plate is captured at many random sub-pixel offsets and at every station of the Z ladder. Sensed area depends on where a feature's edge falls relative to the pixel grid and the projector dots, and on the subtended size D_{px} , which the Z sweep varies by a factor of 4 for each feature. The procedure averages over the phase and also measures its spread.

1. Build the configuration list: $\{T4, T5\} \times \{GAP_SMALL_MM, GAP_LARGE_MM\} \times$ every station of the Z ladder ($Z_STATION_RATIO$). Center the plate and keep it fronto-parallel. Randomize the order within each mounting, so targets are re-mounted as rarely as possible.
2. At each configuration, capture $PHASE_JITTER_POSES_AREA$ poses. Each has a logged random lateral offset uniform over $\pm PHASE_JITTER_SPAN_PX/2$ in H and V, and gets $FRAMES_PER_AREA_POSE$ frames.
3. Field sub-series. At $Z_REFERENCE_MM$ and $G = GAP_SMALL_MM$, repeat Step 2 for each plate at the four off-axis field positions, using $FIELD_SUBSERIES_POSES_AREA$ poses each.
4. Open-background variant (cutouts, optional). At $Z_REFERENCE_MM$, remove the back plate so that nothing lies within the sensor's range behind the holes. Then repeat Step 2. This separates the sensor's fill-in behavior from reads of a real back surface.
5. Capture drift sentinels on the mounted plate's front surface, as Section 5, Step 3 defines them, every $DRIFT_SENTINEL_INTERVAL_MIN$.